Technical Report

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Content

Abstract

Multi-organ ultrasound image classification and segmentation constitute two of the most critical steps in computer-aided clinical diagnosis. Traditional approaches typically require first segmenting the region of interest (ROI) properly and subsequently extracting effective features for disease classification. However, these methods suffer from high computational complexity and inefficiency, rendering them inadequate for multi-organ classification and segmentation tasks. To address these limitations, this paper proposes a prompt-based Boundary-Context Refinement Network (BCR-Net) for multi-organ ultrasound images, which enables efficient simultaneous classification and segmentation. We introduce a Boundary-Context Refinement (BCR) Unit that leverages contextual information from high-level features to enhance object details—particularly in boundary regions—within current-layer features through a reverse attention mechanism.

Method

The overall architecture of the proposed model, as illustrated in Fig.1, comprises four core components: segmentation encoder, segmentation decoder, classification encoder and a classification decoder. Furthermore, to enhance the segmentation decoder's capacity for preserving boundary details, we design the Boundary-Context Refinement (BCR) module, which leverages a reverse attention mechanism to fuse region-level information from global maps with boundary cues from high-resolution features.

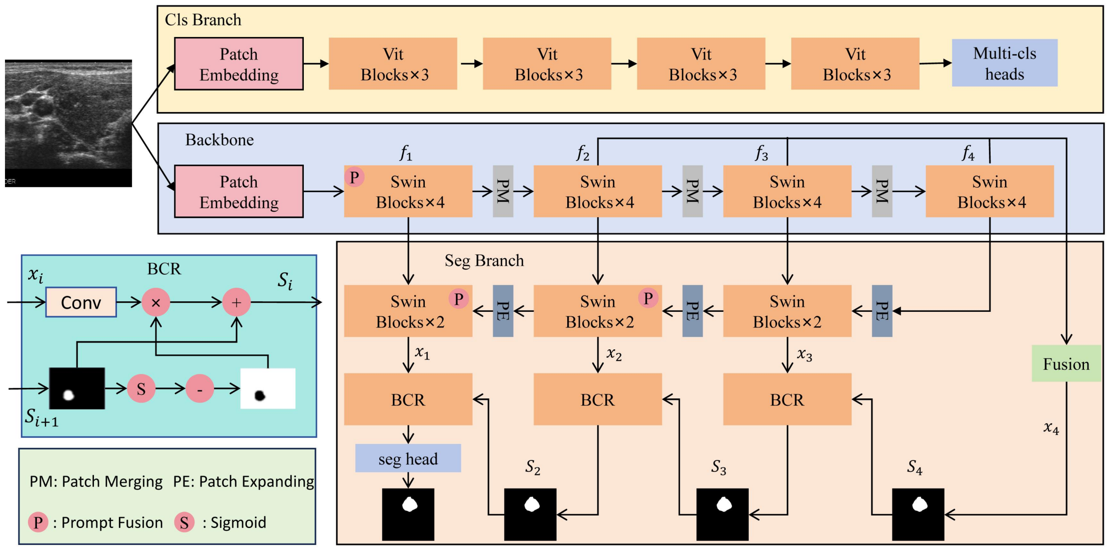

Fig.1. Overview of the proposed BCR-Net.

1) **Segmentation Encoder**. The segmentation encoder addresses limitations of conventional patch embedding modules that employ a single non-overlapping convolutional layer with a large stride (stride=patch_size=4) for initial downsampling—an approach deemed overly aggressive as it prematurely aggregates or discards high-frequency details critical for segmenting anatomical boundaries and lesion contours in medical imaging, while simultaneously fracturing spatial continuity between patches. To resolve these issues, we implement a progressive downsampling scheme using two convolutional layers with overlapping receptive fields: an initial 7×7 convolution (stride=2) captures broader local context, followed by a 3×3 convolution (stride=2) for refined feature processing and secondary downsampling, thereby better preserving spatial structures. This is succeeded by feature extraction through four-stage Swin Transformer blocks, with hierarchical outputs from the last three stages recorded as multi-scale feature maps [f_2, f_3, f_4].

2) **Segmentation Decoder**. The segmentation decoder adopts a UNet-like architecture with skip connections that transfer multi-level features from the shared encoder for concatenation. While deep-layer features contain rich semantic information, they lack boundary detail—a critical factor in medical segmentation. To mitigate this limitation,

we design the Boundary-Context Refinement (BCR) module to enhance feature representation. As shown in Fig.1, the BCR module generates reverse attention feature maps S_i through the following operation:

$$S_i = x_i \odot A_i + S_{i+1},$$
$$A_i = 1 - (\sigma(up(S_{i+1}))), \tag{1}$$

where $\odot$ represents element-wise multiplication, σ is the sigmoid activation function, $up(\cdot)$ denotes the upsampling operation. We fuse the high-level features from the last three layers of the shared encoder to obtain S4=Fusion(f2,f3,f4).

3) **Classification Encoder**. Subsequent experiments reveal that segmentation and classification tasks resist collaborative optimization during training, as the encoder exhibits a bias toward optimizing spatial features critical for segmentation while suppressing abstract semantic representations essential for classification. To resolve this conflict, we design a dedicated classifier encoder branch structured with four stage vision transformer (ViT) blocks. This encoder architecture mirrors the standard ViT-base configuration (12 transformer layers, 768 hidden dimensions) but operates independently from the segmentation pathway. During training, we incorporated a [CLS] token for classification tasks.

4) **Classification Decoder**. The classification dataset comprises binary and four-class breast lesion classification, binary fatty liver classification, and binary appendicitis classification. To address the distinct pathological characteristics across these medical imaging tasks, we employ four dedicated classification heads. This design prevents overlapping representations that cause inter-task interference and tailors structural inductiveness to each disease's diagnostic requirements — enhancing multitask generalization while enabling modular extendability and flexible clinical deployment. Consequently, it substantially optimizes the precision and efficiency of multi-disease diagnostic systems.

5) **Prompt Module**. Effectively learning distinct organ representations from multi-organ datasets presents significant challenges. To address this, we generate organ-specific prompts using one-hot encodings and integrate them into the model via cross-attention mechanisms. These task-adaptive prompts explicitly guide the segmentation process across different anatomical structures. Crucially, prompt fusion occurs at three strategic locations: (1) immediately after patch embedding in the encoder, and (2) within the last two Swin Transformer blocks of both the classification and segmentation decoders. This multi-stage attention-driven integration enables dynamic feature refinement, where organ-prompted contextual information actively steers feature learning. The attention computation is formulated as follows:

$$f_{feature} = Attention(Q_{feature}, K_{prompt}, V_{prompt}) \tag{2}$$

Within this formulation, task designates either classification or segmentation, with $Q_{feature}, K_{prompt}, V_{prompt}$ corresponding to query vectors from the encoder's output features and key/value vectors from the prompt information respectively.

5) Loss Function. For the segmentation task, we adopt a composite loss function combining Cross-Entropy loss and Dice loss:

$$Loss = 0.4 * CE(pred, label) + 0.6 * Dice(pred, label). \quad (3)$$

For classification tasks, direct model training on class-imbalanced data biases learning toward majority classes. To address this issue, we employ Focal Loss as the objective function. Focal Loss mitigates class imbalance by dynamically adjusting sample weights, with its core design suppressing the loss contribution from easy-to-classify examples while amplifying the importance of hard samples and minority classes. Class weights are computed based on class frequencies, assigning higher weights to underrepresented classes—this informs the tuning of hyperparameter α. The hyperparameter γ is set to 2.0. The formulation of Focal Loss is given by:

$$FL(p_t) = -\alpha_t(1 - p_t)^{\gamma}\log(p_t) \quad (4)$$

Experimental Details

The challenge cohort incorporates ultrasound datasets from nine public sources spanning seven anatomic regions, partitioned into 80% training and 20% validation sets while employing a dual sampling strategy to counterbalance both inter-dataset disparities and intra-dataset class imbalances by adaptively increasing sampling frequency for underrepresented datasets and minority classes. Given limited medical data availability, comprehensive augmentation—including standardization, random horizontal flipping, rotation, scaling, and crop/padding—was applied to enhance diversity. For the segmentation model, we trained the entire network for 150 epochs. For the classification model, we froze the segmentation encoder and sequentially fine-tuned the corresponding organ-specific classifiers, training each classifier for 50 epochs. All experiments executed on a single NVIDIA RTX 4090 GPU used batch size 32 with a 0.003 initial learning rate. For the segmentation task, we initialize the segmentation encoder using pre-trained Swin-tiny[1] weights and subsequently train the entire segmentation network. For the classification task, due to the limited data volume, we initialize the classification encoder with the pre-trained USFM[2] model, freeze the encoder during subsequent training, and only finetune the classification heads for different tasks.

Reference

[1] Liu Z, Lin Y, Cao Y, et al. Swin transformer: Hierarchical vision transformer using shifted windows[C]//Proceedings of the IEEE/CVF international conference on computer vision. 2021: 10012-10022.

[2] Jiao J, Zhou J, Li X, et al. Usfm: A universal ultrasound foundation model generalized to tasks and organs towards label efficient image analysis[J]. Medical image analysis, 2024, 96: 103202.